

A Hybrid LoRa–RF Mesh Rover for Post-Disaster Human Detection

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Abstract— The paper proposes a design and implementation of a swarm of rovers for data collection and sharing the same after disasters such as earthquakes and landslides. During these situations, conventional communication infrastructures, including cellular networks and satellite systems become unreliable and a bad option due to physical damage and harsh climatic conditions. However, a decentralized communication architecture based on a LoRa-equipped RF mesh network is proposed. This developed swarm of rovers establishes a reliable communication framework that is independent of communication infrastructures which enables reliable long-range data transmission between these rovers. The rover was enabled with sensors for real-time data-acquisition and machine learning equipped vision module capable of detecting human casualties in regions where human intervention is quite difficult. Multiple microcontrollers are employed to enhance communication efficiency and layered task execution. A webpage serves as an interface which receives and displays the critical messages from the rover to the rescuer side. Experimental validation demonstrates the feasibility of deploying co-operative rover networks for communication and intelligent search operations in disaster-affected areas.

Keywords— *Swarm robotics, disaster management, LoRa mesh network, autonomous rovers, machine learning vision.*

I. INTRODUCTION

In the current premises, natural calamities such as earthquakes and landslides mainly result in great destruction, immense loss of lives [1], [6]. Apart from the human losses, it also causes a rift in the infrastructures of the major ventures worldwide, causing a huge decline in their growth [3]. Citing the recent events that have occurred in countries present in Asia and other neighboring countries in the world, the utmost difficulty faced by the rescue teams were majorly due to heavy dependence of traditional methods of communication systems [2]. Due to the lack of accessibility of these systems in affected areas, it eventually led to the failure of mission in many cases [2]. Due to the lack of proper channel of communication, it clearly disturbs the timely co-ordination between the victims and rescuers, leading to major accidents [1], [3], [6].

Considering the following instances, the primary need of the system is found to be ideal communication for the appropriate management of a disaster and crisis situation [2]. During this timeframe, the cellular network, satellite and internet are

highly affected by the harsh environmental conditions or any physical damage caused by the havoc [2]. To overcome these issues, many researchers have come up with new ideas to communicate in emergency situations, like drone-based rescue systems and Mobile-Ad-Hoc-Network (MANET) [4], [6].

In the recent days, the Long-range wireless technology (LoRa) is quite extensively used in the communication and has become one of the primary choice in disaster and crisis situations, because of its reliability and ability to work under low-power mode [2]. Apart from the stated features, the module is capable to have a very long transmission range and is capable to work under very harsh environmental conditions [2]. The main ability of the LoRa is to work as a mesh network to where the system can simultaneously communicate with each other, upon interconnecting with nodes, transferring reliable information with each other [1], [2].

Based on the above-stated developments, the paper mainly focuses on building a model by proposing a method and implementation of a swarm of rovers along with proper integration of LoRa-RF-mesh based network and communication along with post-disaster data collection for smart and proper detection of victims [1], [3], [7]. The current model accounts for the various disadvantages and loopholes of the previous iterations and models, making it to function independently in harsh infrastructure and have the ability to indulge in long-range communication [1], [3].

II. LITERATURE SURVEY

Current advancements in disaster-related response mainly rely on integrating the network, communication and artificial intelligence alongside low-level robots and rovers. These combine together and help in search and rescue operations (SARs) and help in improving them. Multiple research have widely analysed regarding the multi-robot communication and co-ordination to improve the operation in disaster-fed environment. Bravo-Arrabal et al. [1] proposed a multi-robot SAR framework that combines the 5G network in stand-alone mode with the cloud architectures. The proposed system uses hybrid-wireless networks, robot operating system (ROS) infused edge-based devices and multi-access edge computing between rescue rovers and robots in emergency situations. The study mainly focusses on the importance of

communication in low-latency mode to aid during disaster response.

Reliable communication mainly pose as a major challenge in post-disaster scenarios. Conventional network may get damaged in these scenarios. In order to tackle the problem, Syed et al. [2] developed a LoRa-based communication using mesh network. This system was implemented using a group of rovers. The simulation results showed that the mesh-type-networks based on LoRa provide a widely significant communication with a proper range and stable connection with the advantage of working in low-power mode. Such independent systems are huge takeaway in emergency and critical situations where conventional cellular-based networks fail to work.

Proper and efficient data and information routing are very much important for a robot-based disaster management system. Wang et al. [3] introduced an approach using and attribute-driven Internet of Robotic Things (IoRT) framework essential for post-disaster data collection and proper routing of the same. The proposed method was combined alongside Information-Centric Networking (ICN) in order to select the best and the efficient robots specific to the task and event according to their vital parameters such as connectivity, energy-level and proximity of the robots. The results derived through the experiments showed improved rate in the efficiency of information delivery and communication techniques.

Recent studies have aimed at designing highly advanced hybrid-based robotic platforms that is designed to work under harsh environmental conditions. Krupakara et al. [4] proposed a dual-mobility drone-based rover system that integrates both aerial and ground-based exploration. It combines both the rugged structure of rover alongside the support of Un-manned Aerial Vehicle (UAV) with sensors required for efficient object-detection and terrain monitoring. These hybrid systems improves the operational range and coverage equipped with aerial-based mapping for inspection of disaster-affected areas.

Tele-based operations have also been extensively researched to ensure safe and sound operation of robots and rovers in critical situation and hazardous environments. Aguilar and Barrios [5] developed a LoRa-based system for a long-range control and communication of robot during rescue missions. It incorporates a low-power based wide-area communication and a graphical interface to enhance a proper interaction between human and robot. The results confirmed proper, reliable communication between the robot and the base-station under urban conditions.

In addition to mobility and improvements in communication, advanced sensing technologies integrated with rescue robots to improve detection of victims in hard to reach places. Saqib et al. [6] proposed a drone-mounted system of rover featuring a model based on LiDAR, thermal cameras and proper AI-detection algorithms for a structured search and rescue operations. Similarly, Deep Aditya et al. [7] developed a rover with smart features capable of identifying injured

victims using computer-based vision techniques using standard models like MobileNetV2, automating detection of victims and blood presence during disasters.

Wearable electronics and IoT-based systems have been introduced to enhance response and co-ordination. Reed et al. [8] presented a module representing a wearable device providing exceptional features such as GPS system, emergency alerts and haptic feedback system to assist rescue operation teams to assist the local groups, thereby improving situational awareness during the operations at a crisis situation.

Finally, various network optimization techniques were also involved to improve the communication systems in disaster monitoring system. Jahnave et al. [9] explored various machine learning techniques for delivering content ideas, validating that predictive regression based-models like XGBoost can improve latency and the energy efficiency. This helps in facilitating high-volume data transmission in disaster situations.

Although significant progress have been made in improving the communication networks, the mobility of the robots and the improved and intelligent routing and sensing capability, there are various challenges involved in the achieving a reliable method for communication in affected places.

III. SYSTEM OVERVIEW AND DESIGN

The proposed system is based on decentralized architecture for a swarm of rovers [1], [3]. The architecture aims to support reliable communication, collection of data and detection of survivors in post-disaster situations and damaged environmental conditions [2], [3]. The given architecture poses about deploying many autonomous or semi-autonomous ground rovers interconnected using a LoRa-based RF-mesh network, highly based and inspired by the primary principles of Information-Centric Network (ICN), aiming to allow peer-to-peer communication at a much larger range for the deployed units of rovers [1], [2], [3].

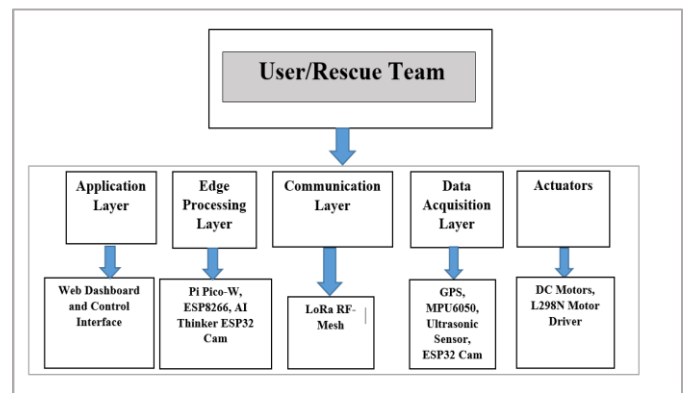


Fig.1 System Architecture of the proposed model

Every single unit of rover in the proposed swarm is individually equipped with basic sensor modules, focussed on real-time data transmission and a machine learning-based computer vision which primarily focusses on detection of

human victims and objects in regions inaccessible for human beings [7]. These swarm of rovers are made to communicate with each other using the sensing modules focussing on situational information for communicating with other rovers and command station using a mesh network [1], [3].

The platform consisting of rovers integrates with multiple controllers in an utmost layered architecture that works towards proper modularity of tasks and improved efficiency [1], [8]. This approach of divide and rule separates the control of the locomotion of the rover, processing of sensor modules, handling of communication, and enabling vision to reduce mishaps to have a smooth operation and co-ordination of the swarm of rovers [8].

Apart from the existing hardware circuitry, for an easy access, a web-based interface is to be developed for the real-time monitoring of the system [5]. Using which the rescue team members can supervise the system from a remote location far from the disaster area, allowing visualization of sensors and proper transmission of data packets [3], [5].

IV. METHODOLOGY

The proposed post-disaster response system is developed using a swarm of rovers based on a cyber-physical framework that is designed to deploy and work in rescue and search operations [1], [3]. The methodology mainly outlines regarding the process of system implementation, design of the system comprising of both the hardware and the software, decentralized communication, remote monitoring, computer-vision-based human detection and validation of the experiment [1], [3], [7]. The block diagram of the dual-rover mesh-based system is given in the Fig.2 below.

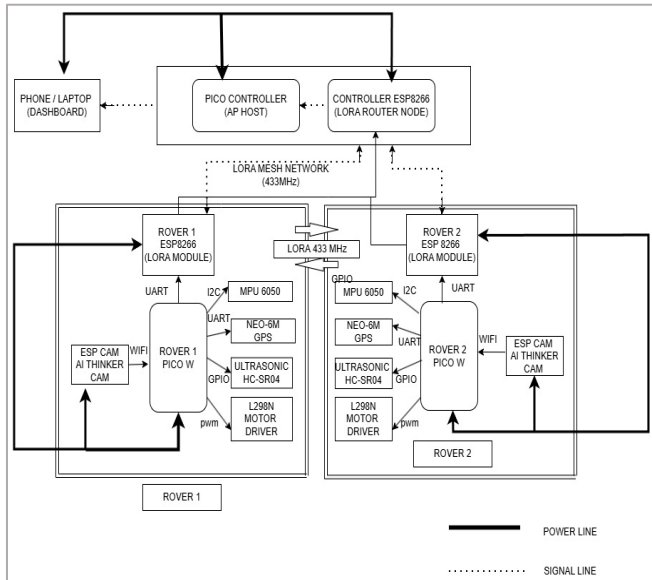


Fig.2 Dual-Rover Mesh Architecture

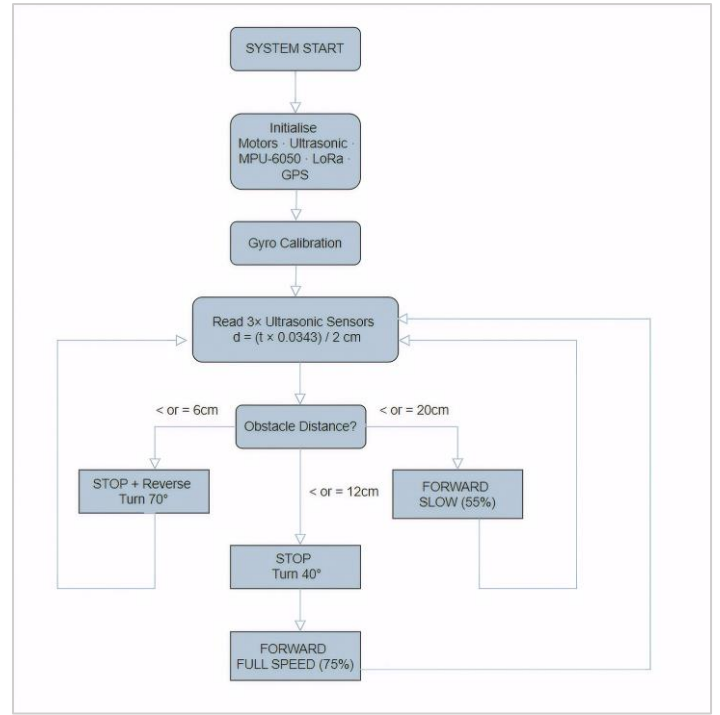


Fig.3 Flowchart for Autonomous Navigation

A. System Design and Swarm Architecture

The system proposed, consists of a multi-ground-level rover units operating autonomously or semi-autonomously in a swarm-based configuration [1], [6]. Each rover unit acts as an independent node within the network, for reliable communication [1], [3]. The Long-range communication between them is achieved using LoRa-enabled RF-mesh-based network [2].

The architecture of the communication is highly inspired by Information-Centric Networking (ICN), where the main aim is to achieve efficient transmission of data packets rather than depending on traditional routing methods [3]. The information generated by an individual rover can be accessed by the control station at any point of time [1], [3]. Due to which, it ensures continuous reliable communication between the rover and the station [1].

The main inspiration of ICN leads the system to focus more on the availability of data rather than routing [3]. This design improves the robustness of the model and makes it fault-tolerant enabling continuous communication, making the system much reliable than usual [1], [3].

B. Hardware Implementation

Each rover platform consists of a number of integrated systems, which is used to cater different functions [8].

Firstly, the processing unit has a layered type of architecture that includes various micro-controllers that cater different functionalities such as locomotion, sensor and vision-based processing [8]. Additionally an ESP32 camera module is used

for object detection and mainly used to acquire visual information and any casualty in the environment [7]. The communication module consists of a LoRa based-transceiver module that purely allows long-range communication even under harsh environmental conditions [2]. The module is also capable to communicate reliably being in a part of a mesh network [1], [2].

The power module consists of rechargeable batteries to power the motor drivers to eventually power up the motor [8]. The batteries are designed in such a way to sustain long periods in the field during the monitoring of the disaster [8]. Power-management becomes a very crucial aspect in post-disaster situation [8].

The hardware design of the rover unit is made in such a way that makes it very much modular in nature [8]. The system is also made to be scalable, reliable and fault-tolerant [1], [8]. These features combine to make sure that the proposed rover unit is made sure to work in very harsh and dynamic environment [1], [8].

C. Machine-Learning-Based Vision Module

The detection capacity of the rover should definitely be made to aid the system [7]. In order to make the best use of that, a computer-vision-based-module has been integrated into the system using machine learning algorithms using software like Edge Impulse [7]. The ESP32 camera module mounted on the rover is used to detect images and object and classify them according to the already fed Machine learning model [7].

The computer-vision module is designed to help a survivor in a post-disaster situation with possible injuries [7]. Once the detection is done, the alert message is transmitted to the rescuer-side using the LoRa communication module [2], [7].

D. Decentralized Data Sharing and Communication

The swarm of rovers is made to establish a network following communication in a decentralized fashion. The network allows for a multi-hop-based-communication, establishing communication from one rover to the monitoring station via the existing nodes [1], [3].

The ICN-inspired communication aims to ensure that the data is transmitted using the various nodes, thereby eliminating the need of a primary base station for the purpose of communication and makes it even reliable. The decentralized approach ensures to have a reliability even in destructed disaster scenario [1], [3].

E. Remote Monitoring and Access

A web-based interface is established for the monitoring of the system to ensure real-time accessing of data from a remote location. This helps in visualizing of the status of the rovers and the network [5], [8].

In addition, the platform enables the operating modules to send alert messages to the rescuer team at the station if

required [5]. This helps in achieving a perfect co-ordination between the swarm and the rescue team.

F. Experimental Validation

The performance and effectiveness of this system are conducted in a very controlled environment [2], [8]. These experiments aims at validating the reliability and the range of the LoRa in the mesh-based network and the stability in the multi-rover system.

Further tests are carried out to make sure that the transmission of data is as accurate to improve the overall robustness of the model. The validation mainly proves the potential of employing such a model in intelligent search and rescue operations in post-disaster situations [1], [3], [6].

V. IMPLEMENTATION

A. Overview of the System

The rover is implemented by using variety of components as mentioned in Table I below. The system is initialized using various protocols involving hardware and software for proper communication. The Raspberry Pi Pico-W is used as the primary logic controller present in each rover module. ESP32 AI-Thinker Camera module is used as the main sensing device which captures and detects the objects. After which, the information is sent to a Pi Pico-W connected to a ESP8266 Node-MCU microcontroller which acts as bridge between LoRa-SX1278 module and Pi Pico-W. Long range transmission capabilities of the LoRa module was properly used for transferring the data-packets from the Pi Pico-W. The Pi Pico-W is made responsible for creating the data packet of a specific format. The header of the data packet states about the type of message transfer. This identifier is essential to process the received data packets and display it in the created webpage-based interface. The system at the receiver side is equipped with LoRa transceiver, which identifies the data sent by individual rover, and differentiate them accordingly to display. The ESP32 AI-Thinker Camera module had a major impact while collecting data.

S.No	Name of the Component
1.	Raspberry Pi Pico W
2.	ESP8266 Node-MCU
3.	ESP32 AI Thinker Camera Module
4.	HC-SR04 Ultrasonic Distance Sensor
5.	MPU 6050 Gyro-Sensor
6.	Neo-6m GPS module
7.	SX-1278 LoRa communication module
8.	18650 Rechargeable Lithium ion battery with Cell Holders
9.	Plastic Chassis Frame for rover with wheels
10.	DC hobby motors
11.	L298N Motor Driver
12.	Jumper Cables and Breadboards

Table I. List of Components

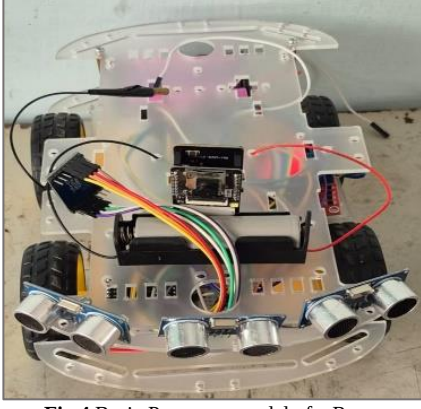


Fig.4 Basic Prototype model of a Rover

B. Hardware Integration

Several hardware components of the rover module involves usage of multiple sensing and control modules which collectively work to ensure autonomous navigation and efficient transmission of data. The Raspberry Pi Pico-W is used as the primary controller to create an interface with neighboring sensors. The L298N Motor Driver module controls the DC motors for locomotion and navigation purposes. The Pulse-Width Modulation (PWM) pulses are calculated using Pico-W to control the speed and motion. Many Ultrasonic Sensors are mounted at critical positions on the rover module to measure the distance between rover and nearby objects. They continuously transmit pulses, helps in calculating the distance based on the de-bounce time. These results helps in identifying a better and a clear pathway. The measurement of distance can be calculated as:

$$d = \frac{t * v_s}{2}$$

Where 'd' is the measured distance (in cm), 't' is the duration of the travelled pulse (in μs) and ' v_s ' represents the speed of sound (in cm/ μs)

The MPU6050 Inertial measurement module is used to track the axis-based motion of the rover. The gyroscope provides the rate of rotation with respect to specific axis. The rate of conversion can be computed as follows:

$$\omega = \frac{R_g}{S_g}$$

Where ' ω ' is the angular rate (in deg/second), ' R_g ' is the raw ADC output and ' S_g ' represents the sensitivity scale factor (in deg/second)

This data is used properly to control the rover at hard turns. The GPS-Module NEO-6M provides the latitude and longitude information enabling rover to send relevant data-packets to the home station.

C. Communication Interface

The main communication interface between rover and the sensor modules is done using UART protocol. The Pi Pico-

W communicates with the ESP8266 Node-MCU module using the UART protocol. It acts as a communication bridge between the rover and LoRa SX1278 module. The approximate time-on-air estimation for LoRa is as follows:

$$T_{packet} = \frac{(N_{payload} + N_{overhead}) * 2^{SF}}{BW}$$

Where ' T_{packet} ' is the time duration of the transmission of the packet (in seconds), ' $N_{payload}$ ' is the size of the payload (in bytes), ' $N_{overhead}$ ' is the protocol of LoRa in overhead and 'SF' refers to the spreading factor and 'BW' represents bandwidth.

The received data at the rover is made to reach a specific format for transmission by the Pico-W. These contain various information regarding the location of rover, sensor information and co-ordinates. Then, these are transmitted via ESP8266 Node-MCU module using the LoRa communication interface. Due to this, the information can be transmitted for a higher range of distance. The LoRa at the receiver's end gets the data, collects it and helps in the proper visualization.

D. Autonomous Navigation

The autonomous navigation of the rover module is done using an algorithm which involves a rule-based algorithm disguised in the form of a Finite State Machine (FSM). Due to which, the rover can move linearly, detect any objects if required and scan within the radius limit.

In the normal mode, it moves forward continuously with the help of the Ultrasonic Sensor. So, when an object is detected, it stops for a while and scans the surroundings to find a clearance path for the rover to commute and also detects objects. After which it resumes with the previous flow of movement. Due to this existing algorithm, it can move in a harsh-environmental condition without the help of human.

E. Data Transmission and Monitoring

The rover module gets the data which contains the sensor information and the status of the system. They are transmitted using a proper communication module. At the receiver's side the status of rover, GPS information and the real-time values are displayed on an interface for monitoring and analyzing the performance of the system.

VI. RESULT

The evaluation of the proposed model were carried out by conducting various tests. The verification of the machine-learning model, object-detection and reliable real-time data transfer and communication were assessed to ensure a reliable and a robust rover-based monitoring system is established. The test results were obtained by testing the ESP32 Camera module mounted on the rover and the integration of the ML model delivered the presence of human.

In addition to this, the communication through LoRa-RF-Mesh based setup to transmit the GPS Co-ordinates and message of detection to the rescue team were tested

A. Feature Classification of the ML Model

The ML model for the proposed classes ‘Fire’ and ‘Human’ were carried out in Edge Impulse platform. The dataset and the hyper-parameter tuning of the datas were initialized in a proper manner to get an optimized result. The distributed features in the explorer window extracted during the training period allowed for analyzing the performance of the ML model.

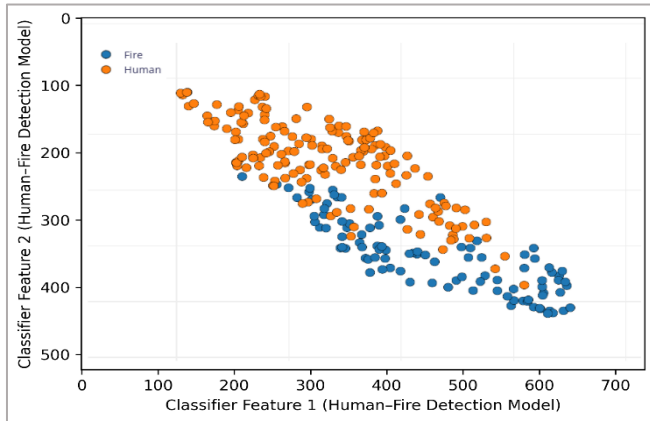


Fig.5 Scattered plot showing the features of human and fire classes in the trained ML model

The above Fig.5 shows the distribution of human and fire classes as a form of a scatter plot. This shows the distribution of the feature space in the window. The classes almost are seen to form an individual cluster between the two classes. Due to this separation of cluster, the ML model trained in a better way to differentiate between human detection and presence of fire and confirming accurate classification of the different classes.

B. Real-Time Object Detection using ESP32-Cam Module

The trained ML model was able to successfully integrate with the AI-Thinker ESP32-Camera Module to test the efficiency of the trained model by implementing real-time object detection and test its capability in real-case test scenarios. The ESP32-Cam module, mounted on the rover uses the deployed object-detection algorithm to detect the objects.



Fig.6 Live Human Detection using ESP32 Cam mounted-Rover



Fig.7 Live Detection of fire using ESP32 Cam mounted-Rover

```
11:02:10.003 -> Object detection bounding boxes:
11:02:10.003 -> Human (0.750000) [ x: 56, y: 40, width: 8, height: 8 ]
11:02:10.842 -> Predictions (DSP: 7 ms., Classification: 694 ms., Anomaly: 0 ms.):
11:02:10.842 -> Object detection bounding boxes:
11:02:10.842 -> Human (0.804688) [ x: 40, y: 40, width: 8, height: 8 ]
11:02:10.842 -> Human (0.937500) [ x: 64, y: 48, width: 8, height: 16 ]
```

```
Predictions (DSP:7ms Class:727ms Anom:0ms):
Fire (0.91) [x:48 y:24 w:16 h:8]
Predictions (DSP:7ms Class:727ms Anom:0ms):
Fire (0.96) [x:56 y:40 w:8 h:24]
Predictions (DSP:7ms Class:727ms Anom:0ms):
Fire (0.88) [x:48 y:48 w:16 h:16]
Predictions (DSP:7ms Class:728ms Anom:0ms):
```

Fig.8 Results of Real-Time Object Detection using the ESP32-Cam Module

The Real-time object detection by the ESP32-Cam module using the detection algorithms is visible in the above Fig.8. The detection system use a bounding-box based detection in order to detect the human present in the frame. The second image clearly depicts the serial-output log which clearly shows the successful detection of human with an impressive F1 Score.

C. Data-Transmission of the Detected Data

After detection of object, obtained from the ESP32-Cam module, the rover creates a detection message with the details of the event and also attaches the GPS Co-ordinates pertaining to the detection event.

```
[Rover] HUMAN DETECTED
[Rover] Lat : 10.9042
[Rover] Lon : 76.8981
[Rover] Sent : <MSG|R1|HUMAN_DETECTED|LAT:10.90420|LON:76.89810>
[Rover]
[Rover] HUMAN DETECTED
[Rover] Lat : 10.9042
[Rover] Lon : 76.8982
[Rover] Sent : <MSG|R1|HUMAN_DETECTED|LAT:10.90420|LON:76.89820>
[Rover]
```

Fig.9 Output Log indicating human-detection with the GPS Co-ordinates transmitted from the rover module

A typical message of the rover module contains the latitude and longitude of the event of detection and the alert message as displayed in the Fig.9. The location of the detected human is successfully communicated later to the ground station using high standard wireless-based communication.

D. Reception of Data

The message detected at the rover module is made to transmit to the ground station using a communication module. The mode of communication used in this system is LoRa, due to its capability of high-range wireless transmission and less power consumption.

```
via LoRa: <MSG|R1|HUMAN_DETECTED|LAT:10.90420|LON:76.89810>
via LoRa: <MSG|R1|HUMAN_DETECTED|LAT:10.90420|LON:76.89810>
via LoRa: <MSG|R1|HUMAN_DETECTED|LAT:10.90420|LON:76.89820>
```

Fig 10. Received message indicating human detection and GPS Co-ordinates received using LoRa module

Fig.10 shows the received message of human detection with the proper Latitude and Longitude information from the rover module using LoRa Communication module. The successful reception of data at the receiver's end verifies the operation and reliability of the system's channel of communication. This communication mechanism ensures that the located individual can be communicated to the control station in real-time.

E. Ground Station and Monitoring Interface

An interface was established to view the incoming alert messages and the status of the rovers at the ground station. This was done to establish real-time monitoring of the system and for ease of access. The interface shows the live telemetry and also contains various information regarding the rover modules and the corresponding sensing units pertaining to it. The detected messages are sent to the people present at the rescuer-side from the base-station.

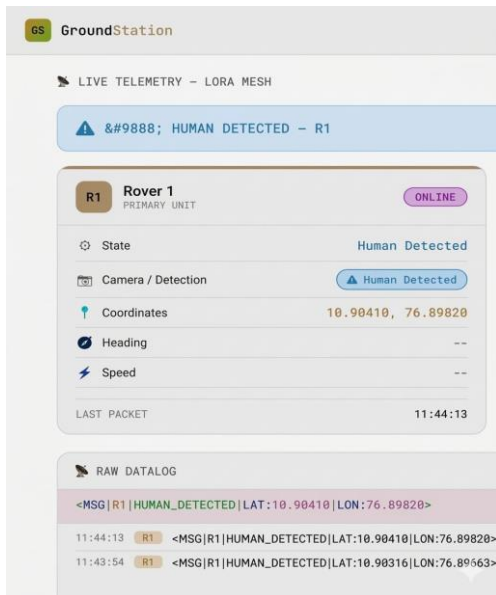


Fig 11. Alert message received in the ground station

The Fig.11 clearly depicts the dashboard of the ground station which shows the current status of the rover module, human detection alert message and notification. This rover telemetry data and the interface helps the members at the rescue team to make proper decisions and give an edge to get the best out of the situation.

VI. CONCLUSION AND FUTURE SCOPE

To conclude, the proposed hybrid LoRa-RF mesh system has a practical approach in cases of disaster situations. The swarm based rovers enabled with LoRa communication, object detection using trained ML models and real-time data monitoring were implemented and viewed in the dashboard. In overall, this system acts as a robust model to implement in a post-disaster scenario.

In the future, the above proposed system incorporated with LoRa-RF-mesh network based system can have an add-on of very advanced highly autonomous navigation and mapping-support system [1], [2]. Methods like Simultaneous Localization and Mapping (SLAM) can be implemented for advanced navigation and routing. Adding to this, aerial based drones can also be added alongside the proposed rovers to improve the operational coverage and efficiency of the rover [4], [6]. Extensive real-time simulation and testing can improve the system [1], [8].

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